Name	ID	Mark
KEY		/100

α=**Sum** of the indicated digits of your ID number

1) For the circuit below, calculate the power delivered to each diode and record in the table. For both diodes $V_{DO} = 0.7$ V. Show all steps.

V_A	R	P_{D1} (mW)	P_{D2} (mW)
+10 V	α kΩ	$3.72/\alpha$	0
-10 V	α kΩ	0	$6.51/\alpha$

D_1 D_2

ON OFF

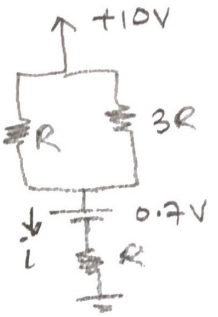
OFF ON

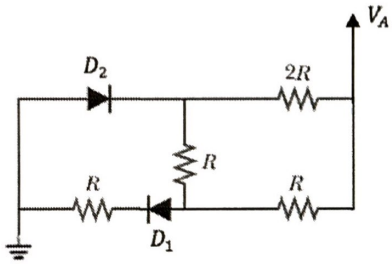
$V_A = +10V$

$i = \frac{10-0.7}{R + R\|3R}$

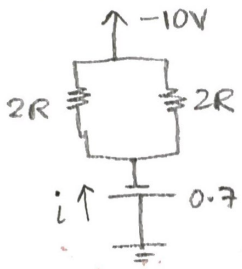
$P_{D1} = i V_{DO}$

$= 3.72/\alpha$





$V_A = -10V$



$i = \frac{10-0.7}{2R\|2R} = 9.3/R$

$P_{D2} = i V_{DO}$

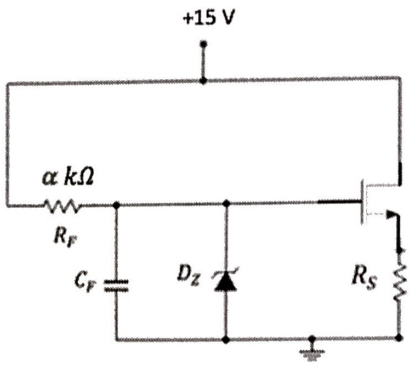
$= 6.51/\alpha$

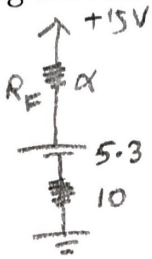
2) The circuit below has been connected for a long time. The Zener diode has $V_{zo} = 5.3$ V, $r_z = 10\ \Omega$ and $I_{z_min} = 0.5$ mA. The NMOS has $V_t = 1$ V, k=1 mA/V², and λ=0. Take $R_S = 1\ k\Omega$ and $C_F = 100\ \mu F$.

a) Calculate the current in resistor R_F.

b) Calculate the current in resistor R_S.

c) What is the effect of doubling the value of R_F on the operation of the circuit? Show.





a) $i_{R_F} = \frac{15-5.3}{\alpha+10} = \frac{9.7}{\alpha+10}$

$V_G = 5.3 + 10(i_{R_F}) \approx 5.3$ V

b) Check mode $V_D > V_G - V_t$

$15 > 5.3 - 1 \Rightarrow$ Sat

$i_{R_S} = k(V_G - V_S - 1)^2$

$= 1m(5.3 - i_{R_S} - 1)^2$

$\Rightarrow i_{R_S}^2 - 9.6\, i_{R_S} + 18.49 = 0 \Rightarrow i_{R_S} = 2.7$ mA

c) If R_F in increased, there is a possibility that the current available for the zener diode becomes less than $I_{z\,min}$. In this case the zener diode will be off and $V_G = 15$ V which will make i_{R_S} higher. The maximum R_F is obtained from $\frac{15 - 5.3}{R_F + 10} = 0.5$ mA

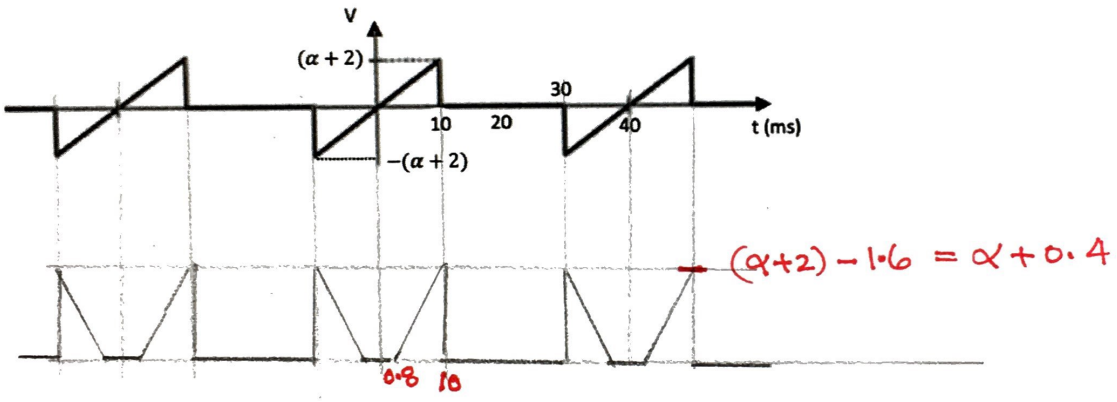
$\Rightarrow R_{F\,max} = 19.4$ kΩ

3) Consider the standard bridge rectifier with a load resistor of 2 kΩ connected to it. The input to the rectifier is the signal shown below. For all diodes $V_{DO} = 0.8$ V.

a) Draw the rectified output voltage waveform. Label all important points.

b) Find the average rectified output voltage and the average load current.

c) If a capacitor of 470 μF is connected to the rectifier, draw the output waveform for this case (on top of the figure in part a). Show all details.



b) $v_{o(ave)} = \frac{\text{area}}{\text{period}} = \frac{2 \times \frac{1}{2}(\alpha + 0.4)(10 - 0.8)}{40}$

$= 0.23(\alpha + 0.4)$ V

$i_{L(ave)} = \frac{V_{o\,ave}}{R_L} = 0.115(\alpha + 0.4)$ mA

c) Since $RC \gg \frac{\text{period}}{2}$, the output voltage will be almost constant.

94 ms $\gg 20$ ms

MOSFET Current Equations	Triode	$I_D = k\,[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$	Saturation	$I_D = k\,(V_{GS} - V_t)^2(1 + \lambda V_{DS})$
Diode Current Equation	Forward	$I_D = I_s\,(e^{V_D/nV_T} - 1)$	Roots	$x = (-b \pm \sqrt{b^2 - 4ac})/2a$